$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{\Upsilon_1 k^2}{7} & -\frac{\Upsilon_1 k^2}{7\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{\Upsilon_1 k^2}{7\sqrt{2}} & -\frac{\Upsilon_1 k^2}{14} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\Upsilon_2 k^2}{14} & \frac{\Upsilon_2 k^2}{7\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\Upsilon_2 k^2}{7\sqrt{2}} & \frac{\Upsilon_2 k^2}{7}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{3\Upsilon_3 k^2}{7} & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{3k^2\overset{(4)}{\kappa}_{15}}{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{2}{3\,\mathrm{Det}(1^+)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & \frac{1}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{1}{3\,\mathrm{Det}(1^-)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} & \frac{2}{3\,\mathrm{Det}(1^-)}
\end{array}
$$

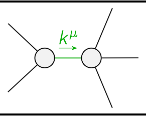
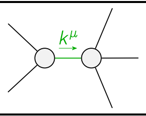
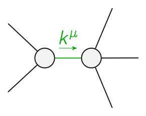
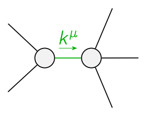
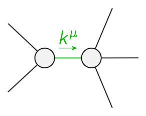
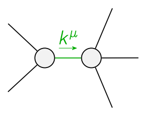
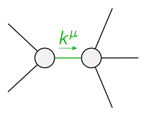
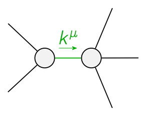
$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{2}{3k^2\overset{(4)}{\kappa}_{15}}
\end{array}
$$

Abbreviations used in matrices

$$\Upsilon_1 == 9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3} \ \&\&\ \Upsilon_2 == 2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} \ \&\&\ \Upsilon_3 == 3\overset{(4)}{\kappa}_{1} + 5\overset{(4)}{\kappa}_{15} \ \&\&$$
$$\mathrm{Det}(1^+) == -\frac{3}{14}k^2(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3}) \ \&\&\ \mathrm{Det}(2^+) == \frac{3}{7}k^2(3\overset{(4)}{\kappa}_{1} + 5\overset{(4)}{\kappa}_{15})$$

Lagrangian

$$\overset{(4)}{\kappa}_{1}\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} - \frac{1}{7}\overset{(4)}{\kappa}_{1}\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \frac{10}{7}\overset{(4)}{\kappa}_{15}\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} +$$
$$\overset{(4)}{\kappa}_{3}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - \frac{18}{7}\overset{(4)}{\kappa}_{1}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + \frac{12}{7}\overset{(4)}{\kappa}_{15}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} -$$
$$2\overset{(4)}{\kappa}_{3}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \frac{18}{7}\overset{(4)}{\kappa}_{1}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \frac{9}{7}\overset{(4)}{\kappa}_{15}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \overset{(4)}{\kappa}_{3}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} +$$
$$\frac{6}{7}\overset{(4)}{\kappa}_{1}\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{18}{7}\overset{(4)}{\kappa}_{15}\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{9}{14}\overset{(4)}{\kappa}_{1}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} -$$
$$\frac{15}{14}\overset{(4)}{\kappa}_{15}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} - \frac{1}{2}\overset{(4)}{\kappa}_{3}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\partial_\delta\mathcal{K}_{\alpha\beta\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{15}\partial_\delta\mathcal{K}_{\beta\alpha\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{0^-} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$	
$\mathcal{J}^{\#1\,\alpha}_{1^-} == \mathcal{J}^{\#2\,\alpha}_{1^-}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$	
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == \mathcal{J}^{\#2\,\alpha\beta}_{1^+}$	3	$3\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$	
Total # constraint(s):	8		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{-8\overset{(4)}{\kappa}_{1}-11\overset{(4)}{\kappa}_{15}}{(2\overset{(4)}{\kappa}_{1}+\overset{(4)}{\kappa}_{15})(3\overset{(4)}{\kappa}_{1}+5\overset{(4)}{\kappa}_{15})}$
	1	0	$-\frac{6\overset{(4)}{\kappa}_{1}+17\overset{(4)}{\kappa}_{15}}{3\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}+5\overset{(4)}{\kappa}_{15}^{2}}$
	1	0	$(36\overset{(4)}{\kappa}_{1}^{3} + 8\overset{(4)}{\kappa}_{15}^{3} + 56\overset{(4)}{\kappa}_{15}^{2}\overset{(4)}{\kappa}_{3} - \sqrt{(2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15})^2(60\overset{(4)}{\kappa}_{1}^{2} + 200\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15} + 281\overset{(4)}{\kappa}_{15}^{2})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3})^2} + 18\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}(5\overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3}) + 2\overset{(4)}{\kappa}_{1}^{2}(83\overset{(4)}{\kappa}_{15} + 14\overset{(4)}{\kappa}_{3}))/(\overset{(4)}{\kappa}_{15}(2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15})(3\overset{(4)}{\kappa}_{1} + 5\overset{(4)}{\kappa}_{15})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3}))$
	1	0	$(36\overset{(4)}{\kappa}_{1}^{3} + 8\overset{(4)}{\kappa}_{15}^{3} + 56\overset{(4)}{\kappa}_{15}^{2}\overset{(4)}{\kappa}_{3} + \sqrt{(2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15})^2(60\overset{(4)}{\kappa}_{1}^{2} + 200\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15} + 281\overset{(4)}{\kappa}_{15}^{2})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3})^2} + 18\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}(5\overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3}) + 2\overset{(4)}{\kappa}_{1}^{2}(83\overset{(4)}{\kappa}_{15} + 14\overset{(4)}{\kappa}_{3}))/(\overset{(4)}{\kappa}_{15}(2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15})(3\overset{(4)}{\kappa}_{1} + 5\overset{(4)}{\kappa}_{15})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3}))$
	1	0	$(135\overset{(4)}{\kappa}_{1}^{3} + 79\overset{(4)}{\kappa}_{15}^{3} + 553\overset{(4)}{\kappa}_{15}^{2}\overset{(4)}{\kappa}_{3} - \sqrt{(225\overset{(4)}{\kappa}_{1}^{4} + 744\overset{(4)}{\kappa}_{1}^{3}\overset{(4)}{\kappa}_{15} + 1342\overset{(4)}{\kappa}_{1}^{2}\overset{(4)}{\kappa}_{15}^{2} + 2440\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}^{3} + 2314\overset{(4)}{\kappa}_{15}^{4})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3})^2} + 3\overset{(4)}{\kappa}_{1}^{2}(281\overset{(4)}{\kappa}_{15} + 35\overset{(4)}{\kappa}_{3}) + \overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}(803\overset{(4)}{\kappa}_{15} + 644\overset{(4)}{\kappa}_{3}))/(\overset{(4)}{\kappa}_{15}(2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15})(3\overset{(4)}{\kappa}_{1} + 5\overset{(4)}{\kappa}_{15})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3}))$
	1	0	$(135\overset{(4)}{\kappa}_{1}^{3} + 79\overset{(4)}{\kappa}_{15}^{3} + 553\overset{(4)}{\kappa}_{15}^{2}\overset{(4)}{\kappa}_{3} + \sqrt{(225\overset{(4)}{\kappa}_{1}^{4} + 744\overset{(4)}{\kappa}_{1}^{3}\overset{(4)}{\kappa}_{15} + 1342\overset{(4)}{\kappa}_{1}^{2}\overset{(4)}{\kappa}_{15}^{2} + 2440\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}^{3} + 2314\overset{(4)}{\kappa}_{15}^{4})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3})^2} + 3\overset{(4)}{\kappa}_{1}^{2}(281\overset{(4)}{\kappa}_{15} + 35\overset{(4)}{\kappa}_{3}) + \overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}(803\overset{(4)}{\kappa}_{15} + 644\overset{(4)}{\kappa}_{3}))/(\overset{(4)}{\kappa}_{15}(2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15})(3\overset{(4)}{\kappa}_{1} + 5\overset{(4)}{\kappa}_{15})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3}))$
	2	0	$-((171\overset{(4)}{\kappa}_{1}^{3} + 458\overset{(4)}{\kappa}_{15}^{3} + 756\overset{(4)}{\kappa}_{15}^{2}\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{1}^{2}(925\overset{(4)}{\kappa}_{15} + 133\overset{(4)}{\kappa}_{3}) + 2\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}(968\overset{(4)}{\kappa}_{15} + 189\overset{(4)}{\kappa}_{3}) + \sqrt{(29241\overset{(4)}{\kappa}_{1}^{6} + 18\overset{(4)}{\kappa}_{1}^{5}(1699\overset{(4)}{\kappa}_{15} + 2527\overset{(4)}{\kappa}_{3}) + \overset{(4)}{\kappa}_{1}^{4}(104773\overset{(4)}{\kappa}_{15}^{2} + 153062\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 17689\overset{(4)}{\kappa}_{3}^{2}) + 4\overset{(4)}{\kappa}_{1}^{3}\overset{(4)}{\kappa}_{15}(138113\overset{(4)}{\kappa}_{15}^{2} + 99638\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 25137\overset{(4)}{\kappa}_{3}^{2}) + 8\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}^{3}(59384\overset{(4)}{\kappa}_{15}^{2} + 106659\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 71442\overset{(4)}{\kappa}_{3}^{2}) + 4\overset{(4)}{\kappa}_{1}^{2}\overset{(4)}{\kappa}_{15}^{2}(286694\overset{(4)}{\kappa}_{15}^{2} + 156394\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 85995\overset{(4)}{\kappa}_{3}^{2}) + 4\overset{(4)}{\kappa}_{15}^{4}(25981\overset{(4)}{\kappa}_{15}^{2} - 12096\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 142884\overset{(4)}{\kappa}_{3}^{2}))})/(\overset{(4)}{\kappa}_{15}(2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15})(3\overset{(4)}{\kappa}_{1} + 5\overset{(4)}{\kappa}_{15})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3})))$
	2	0	$-((171\overset{(4)}{\kappa}_{1}^{3} + 458\overset{(4)}{\kappa}_{15}^{3} + 756\overset{(4)}{\kappa}_{15}^{2}\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{1}^{2}(925\overset{(4)}{\kappa}_{15} + 133\overset{(4)}{\kappa}_{3}) + 2\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}(968\overset{(4)}{\kappa}_{15} + 189\overset{(4)}{\kappa}_{3}) - \sqrt{(29241\overset{(4)}{\kappa}_{1}^{6} + 18\overset{(4)}{\kappa}_{1}^{5}(1699\overset{(4)}{\kappa}_{15} + 2527\overset{(4)}{\kappa}_{3}) + \overset{(4)}{\kappa}_{1}^{4}(104773\overset{(4)}{\kappa}_{15}^{2} + 153062\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 17689\overset{(4)}{\kappa}_{3}^{2}) + 4\overset{(4)}{\kappa}_{1}^{3}\overset{(4)}{\kappa}_{15}(138113\overset{(4)}{\kappa}_{15}^{2} + 99638\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 25137\overset{(4)}{\kappa}_{3}^{2}) + 8\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}^{3}(59384\overset{(4)}{\kappa}_{15}^{2} + 106659\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 71442\overset{(4)}{\kappa}_{3}^{2}) + 4\overset{(4)}{\kappa}_{1}^{2}\overset{(4)}{\kappa}_{15}^{2}(286694\overset{(4)}{\kappa}_{15}^{2} + 156394\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 85995\overset{(4)}{\kappa}_{3}^{2}) + 4\overset{(4)}{\kappa}_{15}^{4}(25981\overset{(4)}{\kappa}_{15}^{2} - 12096\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3} + 142884\overset{(4)}{\kappa}_{3}^{2}))})/(\overset{(4)}{\kappa}_{15}(2\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15})(3\overset{(4)}{\kappa}_{1} + 5\overset{(4)}{\kappa}_{15})(9\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_{3})))$
Resolved unitarity condition(s):	(Demonstrably impossible)		